

Unit 14: Software Process Improvement

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Objective

After studying this unit, you will be able to:

- Discuss approaches to Software process improvement
- Explain maturity models
- Discuss SPI process

Introduction

Software Process Improvement (SPI) is a systematic strategy to improving a software development organization's capacity to build and deliver high-quality software on time and on budget. Software Process, also known as Software Development Process or Software Development Life Cycle (SDLC), is a method of breaking down the software development process into a series of actions or phases that contribute to the creation or development of software. Designing, implementing, changing, testing, and maintaining are examples of these activities.

These activities are carried out in order to improve the project's design and administration. Customizing and improving software processes are other strategies or activities used to improve the overall development process. Software projects especially major software projects are on average 66 percent over budget and 33 percent behind time; up to 17 percent of projects are so terrible that they jeopardize the company's existence.

To address this issue, certain methodologies and strategies for addressing software process concerns, suggesting various improvements and identifying faults and inefficiencies in the process, began to emerge. These strategies have become a standard that firms can use to improve their software development process.

Software process improvement is a method for improving processes. The ultimate goal of software process improvement is to produce a better result than the standard procedure. In previous decades, a crisis occurred from 1965 and 1985. At that time, the rate of project failure was rapidly increasing. The output would not be as good as it needs to be. The crisis manifested in a variety of ways:

1. Project running over budget
2. Project running over time

3. Software was low quality
4. Software often did not meet requirements
5. Project was unmanageable and code difficult to maintain. A lot of problems have come in to the project which leads to not a better result. For example Lack of management, lack of communication among the member of organization. Software process improvement is a set of activities by which processes can achieve a high quality in the organization.

14.1 Software Process Improvement

The Software Process Improvement (SPI) approach is characterised as a set of tasks, tools, and strategies for planning and executing improvement activities with specific goals in mind, such as boosting development speed. Software Process Improvement (SPI) aims to reduce the time, cost, and quality of software engineering and management operations.

SPI activities in software businesses are usually carried out using well-defined reference models such as CMMI and ISO 15504. These models are used to improve development speed, product quality, and cost reduction. SPI is a process re-engineering or change management initiative that identifies inefficiencies in the software development lifecycle and resolves them to improve the process. To be of genuine benefit to the organization, this process should be mapped and connected with organizational goals and change drivers.

A change in the process to produce a better result rather than the original outcome is known as process improvement. A variety of factors influence the software development process. For long-term business, projects should be completed on time, on budget, and with higher quality. It is possible to reuse software process enhancement for the following project. Software process improvement is a set of operations that help organizations mature their processes, achieve all of their objectives, and produce a high-quality output. We can define process improvement as "Software process improvement is a deliberate, planned methodology following standardized documentation practices to capture on paper (and in practice) the activities, methods, practices, and transformations that people use to develop and maintain software and the associated products. As each activity, method, practice and transformation is documented, each is analyzed against the standard of value added to the organization". Software process improvement improves not only quality but also the process of organization.

Software process improvement term is used for improving the organization.

There are some activities for this;

1. Identify the current process
2. Identify the current status of the organization
3. Identify the strength and weakness of the organization
4. Analyse where the changes are needed
5. What will be the effect of these changes?
6. Understand these changes that how much they are affected

Some steps for the process improvement are following:

1. Aware for organization culture
2. Consistence should be there
3. Make target and keep in mind
4. Put a option in the process
5. Set up a well communication team member and senior management
6. Treat process improvement like a project
7. Make changes according to the priority
8. Make a profile of the company
9. Introduce a software process engineering group (SEPG) to your organization
10. Have patience

11. Reuse the previous result

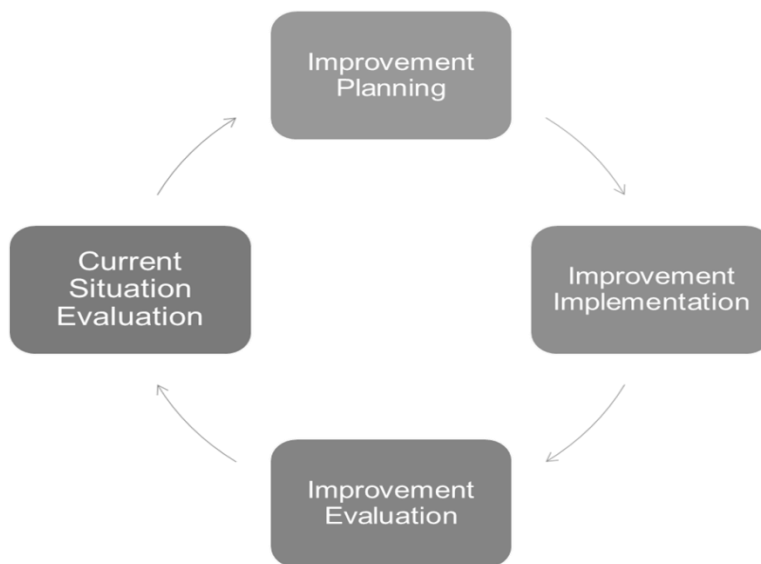
Why do we need for Software Process Improvement?

An immature organization becomes a mature organization as a result of software process improvement. A young company would be unable to provide a high-quality product. There are numerous reasons why high-quality software fails. There is no rationale in the work practices, and there is no support from senior management and management. There is no way to reuse this project because it is just working. The lack of sufficient project documentation renders it unusable for future projects. The project does not meet the needs of the customers. Customer requirements differ from what the project can provide. The project is not completed on schedule or within budget. There are a lot of shortcomings in immature organization. But mature organization provides a good quality to customer with in proper time and budget. There is commitment between the employees. If there is any problem or any problem between management then senior management handle the problem and give a better solution.

14.2 Steps for SPI

- Current Situation Evaluation
- Improvement Planning
- Improvement Implementation
- Improvement Evaluation

Steps for SPI



Current Situation Evaluation

To assess the current situation of the software process by eliciting the requirements from the stakeholders, analyzing the current artefacts and deliverables, and identifying the inefficiencies from the software process.

The elicitation can be conducted through different techniques. For example, individual interviews, group interview, use-case scenarios, and observations. The key considerations in this step to identify organization goals and ask the solution-oriented questions.

Improvement Planning

After analyzing the current situation and the improvement goals, the findings should be categorized and prioritized according to which one is the most important or have the most severity.

In this step, the gap between the current level and the target level should be planned in terms of a set of activities to reach that target. These activities should be prioritized with the alignment of the involved stakeholders and the organization goals.

Improvement Implementation

In this step, the planned activities are executed and it puts the improvements into practice and spreads it across the organization, what can be effective at the 2nd, 3rd, and 4th step that planning and implementation could be an iterative way.

This iterative way of implementation will help the organization to realize the early benefits from the SPI programme early or even adopt the plan if there is no real impact measured from the improvement.

Improvement Evaluation

In this stage measure the changes in case of Improvement. the impact measurement is applied compared with the GQM. The before improvement measures, after the improvement measures, and the target improvement measure. Measurement, in general, permits an organization to compare the rate of actual change against its planned change and allocate resources based on the gaps between actual and expected progress.

Different SPI methods

Similar to the SDLC, SPI has a lot of methods and you can as well define your own method if it is effective or combine between more than one if you do not have any preferences or organization need to adopt a specific method.

SPI methods mainly categorized into two categories: inductive (bottom-up approach) and prescriptive (top-down approach) and most of them are cyclic with four main steps. The table below contains examples of the methods in each category.

SPI methods mainly categorized into two categories:

Inductive (bottom-up approach)

Quality Improvement Paradigm (QIP)

Improvement framework utilizing lightweight assessment and improvement planning (iFLAP)

Prescriptive (top-down approach)

Capability Maturity Model Integration (CMMI)

Software Process Improvement and Capability determination (SPICE)

Advantages of SPI

Standardization and Process consistency: To have a standard and practical process for software development mapped to organization goals and strategy

Cost Reduction: To improve projects cost by enhancing the process and eliminate issues, redundancies, and deficiencies.

Competitive Edge: Being certified in CMMI for example, can put the company in higher competitive edge and make it gain more sales due to the evidence of existing mature software process based on standard method.

Meeting targets and reduce time to market: Meeting organization goals, projects delivery, quality standards, valuable products, professional documentation are outputs from SPI.

Improve customer's satisfaction: Project delivery on time and based on the specification with high quality will improve customer's satisfaction and improve the sales process.

Automation and Autonomy: Introducing tools to automate things and improve quality and ensure consistency. Moreover, enabling different employees to play different roles in the project.

Demotivation factors for SPI

Budget Constraints: SPI is a costly process, because it needs time and dedicated resources, and not only that but also skilled resources especially in SPI. Company may need SPI consultant and train the resources and orient them on SPI initiative.

Time pressure: Due to the nature of the companies to deliver the projects on time, they faced a lot of time pressure which make it harder for them to dedicate time to the SPI project.

Inadequate metrics: Most of the small companies do not have metrics to measure and compare their progress or improvement which make it sometimes impossible to identify measure the improvements of the SPI.

Lack of Management Commitment: It is mainly because the management cannot understand the benefit from SPI and they do not fully support doing this change as well as the other factors like lack of resources, budget, time, etc.

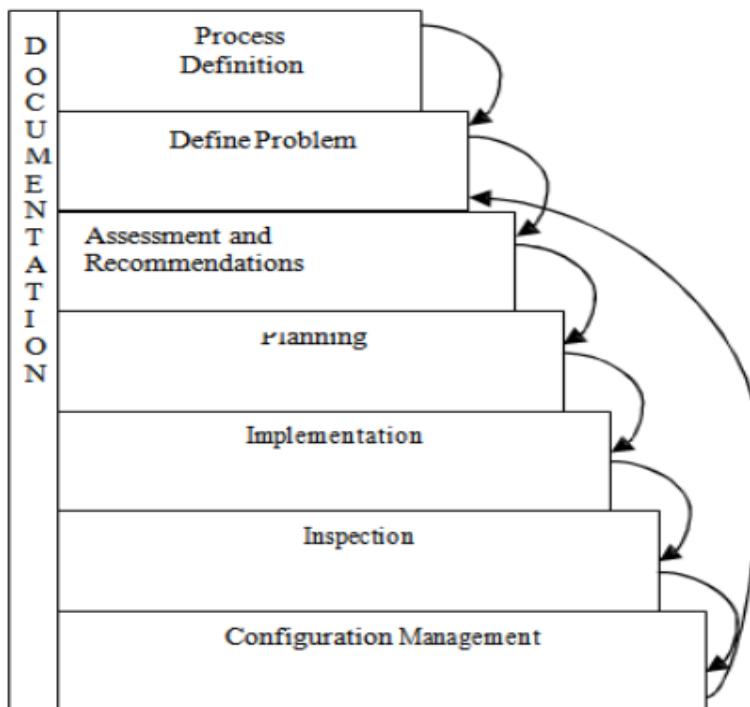
Staff turnover: Sometimes, the company has a high staff turnover which can be an issue to impose the SPI culture change and this can lead to endless SPI.

Micro Organization: Some organization are very small and have very few resources. The SPI will be too big for that kind of companies.

14.3 Software Process Improvement Model (SPIM)

Software process improvement model is an approach or method or both by which it improves process and give better result rather than a normal process. With the software process improvement a better and high quality product could be found within budget and time. The researcher proposes a Software process improvement model in this paper. This model is an iterative as well as cyclic model and it consists of eight steps.

Software process improvement model (SPIM)



Problem in project

There are three problems which are mainly to be solved in the software project.

1. Projects are not of good quality
2. Projects do not deliver in time
3. Project goes over budget

Process Definition and Problem Define

A doctor gives the medicine when he knows about the patient diseases. Without knowing they cannot do anything. Similarly any process improvement model can work accurately if one know the exact problems in the process. Therefore the main issue is to make problem clear at first. When problem become clear then a solution can be found.

A lot of reason may be part of unsuccessful project.

1. Lack of management
2. Lack of right communication inside organization
3. Lack of right communication between customer and organization
4. Technical problem and lack of resources
5. Lack of man power
6. Lack of good training
7. Ego problem
8. Not working in adequate way

Assessment and Recommendation

Once the problem has been described, it is simple to identify areas that need to be improved, and an assessment team is formed based on the problem and the recommendations of higher authorities. The size of the problem and the size of the team are determined by the management. The team should be made up of personnel that are highly qualified and experienced. If the problem is modest, a team of two to three researchers will suffice. If the improvement area is large, a team will be formed based on the problem's size. Sometimes software enhancement is ineffective because no one can claim or guarantee that a methodology or process will improve the project's quality or the organization's culture. So if area of improvement is small like in small project, then risk is reduced. But if area of improvement is of higher level then it may be a great risk. A project/organization take time for improvement and have no guarantee for success, but only probability for the success of same. So at this condition it is better to take sponsorship by the third party. But it requires higher authority's permission. They can take decision about timing of sponsorship needed. They have a meeting with senior management about the improvement of process and assessment of team like who should be the member of team for process improvement. This is very important step because team should be consist of highly qualified member and as small as possible with their role defined and known to them. Member of the team must be certified by SEI (software engineering institute) as only then they can take better decision in critical situation.

Planning

They begin working on the project after the evaluation team has been formed. They devise a strategy for improving the process by identifying improvements that are required. First and foremost, the scope of the problem should be clearly identified. The team is now creating surveys for individuals from various levels inside the business. After analyzing the company's existing state, a profile for the company's future is created. They do this by interviewing as many people as possible and devising a strategy. If there are multiple strategies for improving a process, they should choose the one that yields the best results. With the approach defined, a detailed implementation plan can be developed by taking consideration of all the factors like, technical and nontechnical. When the team complete the plan, they have a meeting with the senior management and there they give a presentation to the seniors. By doing this they reduce risk level. At this situation assessment team is not responsible for any casualty. If the senior management comfortable with your plan, only then team would be allow to move towards next step of implementation plan. If they don't satisfied with plan, then alternative plan have to make along with the presentation. This plan includes schedule, tasks, milestones, decision points, resources, responsibilities, measurement, tracking mechanisms, risks and mitigation strategies, and any other elements required by the organization.

Implementation

When the plan is established in accordance with the problem, the solution is created in accordance with the plan. Make a paper solution first, based on the evaluation team's experience, knowledge, and expertise. After that, he makes another presentation to the higher authorities. If they offer any suggestions for improving the solution, adjust the plan accordingly, and if they are satisfied with the solution, begin implementation. They opt for a better option. The plan should be discussed with top management and the organization before it is implemented. Begin working on the solution once they have reached an agreement. It is a step-by-step method that does not apply to all of the solutions in one go. Makes a baseline after every step and take a pilot test and check the result if the result is positive then next continue the solution. If there are more than one solution. Then one shouldn't take decision by himself in this situation. Again discuss with the higher authorities. Whatever solution seems better, apply the solution. By doing this risk is less because at this

situation team is not responsible and if the plan be successful the all the credit goes to the assessment team.

Inspection and Configuration Management

In most cases, an inspection is a systematic examination or a formal evaluation process. It entails the application of measurements, tests, and gauges to specific properties of an object or activity. The results are frequently compared to pre-determined requirements and standards to see if the item or activity meets these goals. The majority of inspections are nondestructive. The accuracy of the work plan and execution cannot be guaranteed. A team of moderator, reader, and inspectors is assembled for the inspection. There is a formal meeting. The moderator's job is to keep the meeting on track and guarantee that the agenda is followed. Role of the moderator is like a leader of the inspection team. So moderator should be highly qualified and skilful. Moderator tells the team how the inspection starts and leads them. Inspection team takes the interview of assessment team. They make a questionnaire for every member of team.

Review of the work done by all and they analyse the result. What result will come after process improvement? Review all the factors came in the technique. Main motive of the inspection is to find the defect in the module. If there are no defects then team will give the clean chit for the process improvement. If there is any problem in the technique then they tell where the problem is or what problem can be faced. What change should be done in process? They specify where the changes are need. This work is handling by the configuration management. Before a configuration item makes a baseline of the work. A baseline is a software configuration management concept that helps us to control changes without seriously impeding justifiable changes. Once a baseline is established, we pass through a one way door. After that changes inspection team again inspect the process. This process will continue till the defect does not end. And the improvement has done.

Documentation

In this paradigm, documentation is an umbrella activity that occurs concurrently with the process. Each phase necessitates documentation. Documentation is required for future work so that it can be reused and new work can be completed. There are a number of requirements related with the documents linked with a software project and the system being developed:

They should act as a communication medium between members of the development team. They should be a system information repository to be used by maintenance engineers.

Software Process Improvement Models

- CMMI
- SIX SIGMA
- SPICE
- TickIT Guide ISO 9001
- BOOTSTRAP

CMMI

CMMI is a framework that provides a process improvement setup for the software engineering and product development. CMMI combined best practices and basic principles and interlinked them in order to improve the process inside an organization.

CMMI currently addresses three areas of interest:

Product and service development - CMMI for Development (CMMI-DEV)

Service establishment, management, and delivery - CMMI for Services (CMMI-SVC)

Product and service acquisition - CMMI for Acquisition (CMMI-ACQ)

SIX SIGMA

Six Sigma is a process improvement management framework to achieve bottom-line results, and customer's loyalty. The objective of Six Sigma is the implementation of a measurement based strategy that is focused on process improvement and variation reduction.

Six Sigma is a statistically-based process improvement methodology that aims to reduce defects to a rate of 3.4 defects per million defect opportunities by identifying and eliminating causes of variation in business processes. SIX SIGMA is iterative methodology reduced the defect one by one. Six Sigma focuses on prioritizing and solving specific problems which are selected based on the

strategic priorities of the company and the problems which are causing the most defects. SIX SIGMA methodology works on two approaches DMAIC, DMADV.

SPICE

The purpose of SPICE framework is to analyse and assess the software process and on these assessments, provide information about process strengths, weaknesses and capability to achieve its goals. SPICE was inspired by CMM and other software process improvement models.

TickIT Guide ISO 9001

The purpose of stimulating software system developers to think about process quality, how to achieve the desired quality and how to manage this quality in order to perform continuous improvement. TicketIT also provides a practical framework for the management of software development quality.

BOOTSTRAP

BOOTSTRAP is a European method for software process assessment and improvement that was designed and developed to speed up the application of software engineering technology.

14.4 Maturity model

A maturity model is applied within the context of an SPI framework. The intent of the maturity model is to provide an overall indication of the “process maturity” exhibited by a software organization.

- an indication of the quality of the software process, the degree to which practitioner’s understand and apply the process,
- The general state of software engineering practice.

Capability Maturity Model (CMM)

Capability Maturity Model is used as a standard to measure the maturity of an organization's software process. CMM was developed by the Software Engineering Institute (SEI) at Carnegie Mellon University in 1987.

Capability Maturity Model (CMM) specifies an increasing series of levels of a software development organization. The higher the level, the better the software development process.

It isn't a model of a software process. It is a framework for analysing any organization's approach and processes for developing software products. It also includes advice for further improving the maturity of the software development process. It is built on the most effective feedback and growth strategies used by the world's most successful firms.

This model describes a strategy for software process improvement that should be followed by moving through 5 different levels. Each level of maturity shows a process capability level. All the levels except level-1 are further described by Key Process Areas (KPA's).

Key Process Areas (KPA's)

Each of these KPA's defines the basic requirements that should be met by a software process in order to satisfy the KPA and achieve that level of maturity.

Key process areas form the basis for management control of the software project and establish a context in which technical methods are applied, work products like models, documents, data, reports, etc. are produced, milestones are established, quality is ensured and change is properly managed.

CMM Maturity Levels

- Initial
- Repeatable
- Defined
- Initial Managed
- Optimizing

CMM -three principal areas

Organization and resource management. This deals with functional responsibilities, personnel, and other resources and facilities.

Software engineering process and its management. This concerns the scope, depth, and completeness of the software engineering process and the way in which it is measured, managed, and improved.

Tools and technology. This deals with the tools and technologies used in the software engineering process and the effectiveness with which they are applied. This section is not used in maturity evaluations or assessments.

Limitations of CMM Models

- CMM determines what a process should address instead of how it should be implemented.
- It does not explain every possibility of software process improvement.
- CMM will not be useful in the project having a crisis right now.
- It concentrates on software issues but does not consider strategic business planning, adopting technologies, establishing product line and managing human resources.
- It does not tell on what kind of business an organization should be in.

14.5 Maturity models – CMMI

CMMI model is a proven set of best practices organized by critical business capabilities which improve business performance. It is designed to be understandable, accessible, flexible, and integrate with other methodologies such as agile.

CMMI is a framework that provides a process improvement setup for the software engineering and product development. CMMI combined best practices and basic principles and interlinked them in order to improve the process inside an organization.

CMMI highlights

- Reduce rework by 54%.
- Improve product quality up to 70%.
- Improve productivity up to 54%.
- Average improvement in defect containment 23%
- Development velocity increased up to 38%

CMMI V2.0 Key Improvements***Improve Business Performance***

Business goals are tied directly to operations in order to drive measurable, improved performance against time, quality, budget, customer satisfaction and other key drivers.

Build Agile Resiliency and Scale

Direct guidance on how to strengthen agile with Scrum project processes with a focus on performance.

Increase Value of Benchmarking

New performance-oriented appraisal method improves reliability and consistency of benchmarks while reducing preparation time and lifecycle costs.

Accelerate Adoption

Online access and adoption guidance make the benefits of CMMI more accessible than ever.

14.6 Appraisal

An appraisal is an activity that helps you to identify the strengths and weaknesses of your organization's processes and to examine how closely the processes relate to CMMI best practices.

Need of Appraisals

Determine how well the organization's processes compare to CMMI best practices, and identify areas where improvement can be made. Inform external customers and suppliers about how well the organization's processes compare to CMMI best practices. Meet customer contractual requirements.

CMMI Appraisal Method

The CMMI appraisal method is used to identify process implementation strengths and weaknesses as well as process persistence and habit.

The method is also used to link demonstrated business performance relative to the model adoption

Standard CMMI Appraisal Method for Process Improvement (SCAMPI)

There are three appraisal classes: Class A, B and C.

SCAMPI A: The most rigorous appraisal method, SCAMPI A is most useful after multiple processes have been implemented. It provides a benchmark for businesses and is the only level that results in an official rating.

SCAMPI B: This appraisal is less formal than SCAMPI A; it helps find a target CMMI maturity level, predict success for evaluated practices and give the business a better idea of where they stand in the maturity process.

SCAMPI C: This appraisal method is shorter, more flexible and less expensive than Class A or B. It's designed to quickly assess a business's established practices and how those will integrate or align with CMMI practices. It can be used at a high-level or micro-level, to address organizational issues or smaller process or departmental issues.

The new appraisal method is capable of supporting appraisals in a variety of contexts:

- Benchmarking
- Internal performance and process improvement
- Process monitoring
- Supplier selection
- Risk reduction

14.7 CMMI Maturity Levels

- Initial
- Repeatable
- Defined
- Initial Managed
- Optimizing

Initial: Processes are viewed as unpredictable and reactive. At this stage, "work gets completed but it's often delayed and over budget." This is the worst stage a business can find itself in - an unpredictable environment that increases risk and inefficiency.

Managed: There's a level of project management achieved. Projects are "planned, performed, measured and controlled" at this level, but there are still a lot of issues to address.

Defined: At this stage, organizations are more proactive than reactive. There's a set of "organization-wide standards" to "provide guidance across projects, programs and portfolios." Businesses understand their shortcomings, how to address them and what the goal is for improvement.

Quantitatively managed: This stage is more measured and controlled. The organization is working off quantitative data to determine predictable processes that align with stakeholder needs. The business is ahead of risks, with more data-driven insight into process deficiencies.

Optimizing: Here, an organization's processes are stable and flexible. At this final stage, an organization will be in constant state of improving and responding to changes or other

opportunities. The organization is stable, which allows for more “agility and innovation,” in a predictable environment.

CMMI model components

Process areas

24 process areas that are relevant to process capability and improvement are identified. These are organized into 4 groups.

Goals

Goals are descriptions of desirable organizational states. Each process area has associated goals.

Practices

Practices are ways of achieving a goal - however, they are advisory and other approaches to achieve the goal may be used.

CMMI Process Area

A Process Area is a cluster of related practices in an area that, when implemented collectively, satisfy a set of goals considered important for making significant improvement in that area. All CMMI process areas are common to both continuous and staged representations.

The CMMI Process Areas PAs can be grouped into the four categories to understand their interactions and links with one another regardless of their defined level

- Process Management
- Project Management
- Engineering
- Support

Each process area is defined by a set of goals and practices. Generic goals and practices: They are part of every process area. Specific goals and practices: They are specific to a given process area. There are 22 process areas in CMMI that indicate the various aspects of product development.

Project Management

Integrated Project Management (IPM)

Project Monitoring and Control (PMC)

Project Planning (PP)

Quantitative Project Management (QPM)

Requirements Management (REQM)

Risk Management (RSKM)

Supplier Agreement Management (SAM)

Engineering

Product Integration (PI)

Requirements Development (RD)

Technical Solution (TS)

Validation (VAL)

Verification (VER)

Process Management

Organizational Performance Management (OPM)

Organizational Process Definition (OPD)

Organizational Process Focus (OPF)

Organizational Process Performance (OPP)

Organizational Training (OT)

Support

Casual Analysis and Resolution (CAR)

Configuration Management (CM)

Decision Analysis and Resolution (DAR)

Measurement and Analysis (MA)

Process and Product Quality Assurance (PPQA)

Summary

Software Process Improvement (SPI) methodology is defined as definitions of sequence of tasks, tools and techniques to be performed to plan and implement improvement activities.

SPI mainly consists of 4 cyclic steps:

Current Situation Evaluation

Improvement Planning

Improvement Implementation

Improvement Evaluation

CMM means Capability Maturity Model. It is a series standard to assess the software capability and maturity. It also provides the methodologies for software assessment.

Capability Maturity Model Integration (CMMI) is a successor of CMM and is a more evolved model that incorporates best components of individual disciplines of CMM like Software CMM, Systems Engineering CMM, People CMM, etc.

Keywords

SPI CMM

CMMI SIX SIGMA

SPICE ISO

SDLC BOOTSTRAP

QIP

Self Assessment

1. What are the steps for Software process improvement?
 - A. Current Situation Evaluation
 - B. Improvement Planning
 - C. Improvement Implementation
 - D. Above all

2. Software process improvement methods are__
 - A. Inductive
 - B. Prescriptive
 - C. Both inductive and prescriptive
 - D. None of above

3. What are the advantages of Software process improvement?
 - A. Cost reduction
 - B. Competitive Edge

-
- C. Standardization and Process consistency
 - D. All of above
4. What are the Demotivation factors for Software process improvement?
- A. Budget Constraints
 - B. Time pressure
 - C. Inadequate metrics
 - D. All of above
5. CMMI and SPICE are part of__
- A. Prescriptive
 - B. Inductive
 - C. Both inductive and prescriptive
 - D. None of above
6. Which is not Software Process Improvement Model?
- A. SIX SIGMA
 - B. IBM 02
 - C. SPICE
 - D. TickIT Guide ISO 9001
7. What are the areas of interest for CMMI?
- A. Product and service development
 - B. Service establishment, management, and delivery
 - C. Product and service acquisition
 - D. All of above
8. BOOTSTRAP is__
- A. Asian method
 - B. European method
 - C. American method
 - D. All of above
9. What are the principal area of CMM?
- A. Organization and resource management
 - B. Software engineering process and its management
 - C. Tools and technology
 - D. All of above
10. Which is not part of CMM Maturity Level?
- A. Repeatable
 - B. Defined
 - C. Testing
 - D. Initial Managed

11. CMMI V2.0 Key Improvements are__
- A. Improve Business Performance
 - B. Build Agile Resiliency and Scale
 - C. Build Agile Resiliency and Scale
 - D. Above all
12. Which is not types of CMMI Appraisals?
- A. Benchmark Appraisal
 - B. System design appraisal
 - C. Sustainment Appraisal
 - D. Action Plan Reappraisal
13. What are the CMMI Maturity Levels?
- A. Repeatable
 - B. Defined
 - C. Initial Managed
 - D. All of above
14. CMMI model components are__
- A. Process areas
 - B. Goals
 - C. Practices
 - D. All of above
15. Process Management and Project Management are part of__
- A. Practices
 - B. Goals
 - C. CMMI Process Areas
 - D. None of above

Answers for Self Assessment

- | | | | | |
|-------|-------|-------|-------|-------|
| 1. D | 2. B | 3. D | 4. D | 5. A |
| 6. B | 7. D | 8. B | 9. D | 10. C |
| 11. D | 12. B | 13. D | 14. D | 15. C |

Review Questions

1. What Is Maturity Level?
2. Define Is Software Process?
3. Differentiate between CMM and CMMI.
4. What is significance of Software Process Improvement?

5. Discuss Key Process Areas.
6. What are the CMMI model components?
7. Differentiate between SIX SIGMA and SPICE.
8. What is the role of ISO, give a suitable example.



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